

Mathematics for Aerospace Engineers

Chapter 2: The Laplace transform

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The Laplace Transform

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2 The Laplace transform

2.1 Introduction

The Laplace Transform takes a function of a real variable t (usually time), $f(t)$ to a function of a complex variable s (frequency), $\bar{f}(s)$.

It is useful for analysing complicated functions, and is used widely in Science and Engineering, e.g. in Control Theory.



In this chapter, we will use the Laplace Transform as a tool to solve ODEs that are difficult, or even impossible, to solve otherwise.

2.1.1 Solving ODEs

The basic approach is not to solve the ODE for $y(x)$ directly, but to introduce a transform $\mathcal{L}$ to give an algebraic equation for a new variable $\bar{y}(s)$. Having solved for $\bar{y}(s)$, the solution $y(t)$ is obtained by applying the inverse transform $\mathcal{L}^{-1}$.

ODE
for $y(t)$

(i) use transform

→

$$\bar{y} = \mathcal{L}(y)$$

algebraic
equation for
 $\bar{y}(s)$

↓ (ii) solve for
 $\bar{y}(s)$

solution
 $y(t)$

(iii) inverse
transform

←

$$y = \mathcal{L}^{-1}(\bar{y})$$

$$\bar{y} = \bar{y}(s)$$

2.2 Definition and basic properties

Given a function f , defined on the set of all $t \geq 0$, the *Laplace transform* of f , is denoted $\mathcal{L}\{f(t)\}$, or $\bar{f}(s)$, where s is a complex variable in the frequency domain. The Laplace transform is defined as

$$\mathcal{L}\{f(t)\} = \bar{f}(s) = \int_0^{\infty} e^{-st} f(t) dt,$$

provided this integral exists.

Linearity

Since integration is a linear operation, the Laplace transform is a *linear transformation*, i.e. for functions $f_1(t)$, $f_2(t)$ and real constants a and b ,

$$\mathcal{L}\{af_1(t) + bf_2(t)\} = a\mathcal{L}\{f_1(t)\} + b\mathcal{L}\{f_2(t)\}.$$

This important property means that we can take the Laplace (or inverse Laplace) transform of a function $f(t)$ (or $\bar{f}(s)$) term-by-term.

2.2.1 Laplace transform of derivatives

To see why the Laplace transform is useful, let us look at how it transforms derivatives.

$$\begin{aligned}\mathcal{L}\left\{\frac{df}{dt}\right\} &= \int_0^{\infty} e^{-st} \frac{df}{dt} dt, \quad (\text{int. by parts}) \\ &= [e^{-st} f]_0^{\infty} - \int_0^{\infty} (-s) e^{-st} f dt, \\ &= -f(0) + s \underbrace{\int_0^{\infty} e^{-st} f(t) dt}_{\bar{f}(s)}, \\ &= -f(0) + s\bar{f}(s).\end{aligned}$$

$$\begin{aligned}
\mathcal{L}\left\{\frac{d^2f}{dt^2}\right\} &= \int_0^\infty e^{-st} \frac{d^2f}{dt^2} dt, \quad (\text{int. by parts}) \\
&= \left[e^{-st} \frac{df}{dt} \right]_0^\infty - \int_0^\infty (-s) e^{-st} \frac{df}{dt} dt, \\
&= -f'(0) + s \underbrace{\int_0^\infty e^{-st} \frac{df}{dt} dt}_{\mathcal{L}\{f'(t)\} = -f(0) + s\bar{f}(s)}, \\
&= -f'(0) - sf(0) + s^2\bar{f}(s).
\end{aligned}$$

These two results are useful for solving ODEs, as the Laplace transforms include no derivatives of $\bar{f}$. If we apply the Laplace transform to an ODE, we obtain a purely algebraic equation.

The Laplace transform method has two advantages:

- 1 Initial conditions are built in from the start. This means the method is particularly suitable for initial value problems, where $f(0)$ and $f'(0)$ are known.
- 2 There is no need to guess a particular integral.

2.2.2 Laplace transform of an integral

We will not make use of the Laplace transform of an integral in this module, but we note that it is given by the following.

$$\mathcal{L} \left\{ \int_0^t f(k) \, dk \right\} = \int_0^\infty e^{-st} \left[\int_0^t f(k) \, dk \right] dt. \quad (\text{int. by parts})$$

Note that

$$\frac{d}{dt} \left[\int_0^t f(k) \, dk \right] = f(t).$$

Then

$$\begin{aligned} \mathcal{L} \left\{ \int_0^t f(k) \, dk \right\} &= \left[-\frac{1}{s} e^{-st} \int_0^t f(k) \, dk \right]_0^\infty - \int_0^\infty \left(-\frac{1}{s} \right) e^{-st} f(t) \, dt, \\ &= 0 + \frac{1}{s} \underbrace{\int_0^\infty e^{-st} f(t) \, dt}_{\bar{f}(s)} = \frac{\bar{f}(s)}{s}. \end{aligned}$$

2.3 Laplace transform of some important functions

Example 2.3.1

Let $f(t) = 1$. Then

$$\bar{f}(s) = \int_0^{\infty} e^{-st} dt = \left[\frac{e^{-st}}{s} \right]_0^{\infty} = \frac{1}{s},$$

provided $s > 0$ (otherwise the Laplace transform would not exist).

Hence, $\mathcal{L}\{1\} = \frac{1}{s}$.

Example 2.3.2

Let $f(t) = e^{at}$, where a is a real constant. Then

$$\begin{aligned}\bar{f}(s) &= \int_0^{\infty} e^{-st} e^{at} dt, \\ &= \int_0^{\infty} e^{(a-s)t} dt, \\ &= \left[\frac{e^{(a-s)t}}{a-s} \right]_0^{\infty}, \\ &= \frac{1}{s-a},\end{aligned}$$

provided $s > a$ (i.e. $a - s < 0$).

Hence, $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$.

Example 2.3.3

Let $f(t) = \sin(at)$, where a is a real constant. Then

$$\begin{aligned}\bar{f}(s) &= \int_0^{\infty} e^{-st} \sin(at) dt, \quad (\text{int. by parts}) \\ &= \left[\frac{e^{-st} \sin(at)}{-s} \right]_0^{\infty} - \int_0^{\infty} \frac{ae^{-st} \cos(at)}{-s} dt, \\ &= 0 + \frac{a}{s} \int_0^{\infty} e^{-st} \cos(at) dt, \quad (\text{int. by parts again}) \\ &= \frac{a}{s} \left\{ \left[\frac{e^{-st} \cos(at)}{-s} \right]_0^{\infty} - \int_0^{\infty} \frac{-ae^{-st} \sin(at)}{-s} dt \right\}, \\ &= \frac{a}{s^2} - \frac{a^2}{s^2} \bar{f}(s).\end{aligned}$$

Hence, $\left(1 + \frac{a^2}{s^2}\right) \bar{f}(s) = \frac{a}{s^2}$, which gives $\bar{f}(s) = \mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}$.

By performing similar calculations, we build up a table of Laplace transforms (which you will find on Page 3 of the Formula Sheet). Even better, we can then use the table to find *inverse Laplace transforms*.

For each function $\bar{f}(s)$ in the table, there is a corresponding function $f(t)$ such that $\bar{f}(s)$ is the Laplace transform of $f(t)$.

$f(t)$ and $\bar{f}(s)$ are often called *Laplace transform pairs*.

$$\mathcal{L}\{f(t)\} = \bar{f}(s) \quad \text{and} \quad \mathcal{L}^{-1}\{\bar{f}(s)\} = f(t),$$

where $\mathcal{L}^{-1}\{\bar{f}(s)\}$ is the inverse Laplace transform of $\bar{f}(s)$. For example

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \quad \text{and} \quad \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}.$$

Without such tables, finding inverse Laplace transforms would not be easy!

2.4 Application of the Laplace transform to ODEs

Example 2.4.1

A first order ODE is given by

$$\frac{dy}{dt} + y = e^{2t} \quad \text{with} \quad y(0) = 1. \quad (1)$$

There are three stages in the process.

- 1 First take the Laplace transform of both sides of (1),

$$\begin{aligned} \mathcal{L}\left\{\frac{dy}{dt}\right\} + \mathcal{L}\{y\} &= \mathcal{L}\{e^{2t}\}, \\ \Rightarrow s\bar{y}(s) - 1 + \bar{y}(s) &= \frac{1}{s-2} \end{aligned} \quad (2)$$

- 2 Solve the algebraic equation (2) for $\bar{y}(s)$.

$$\bar{y}(s) = \frac{1}{s+1} + \frac{1}{(s+1)(s-2)} = \frac{s-1}{(s+1)(s-2)}. \quad (3)$$

- 3 Apply the inverse Laplace transform to $\bar{y}$ from equation (3). This gives $y(t) = \mathcal{L}^{-1}\{\bar{y}(s)\}$.

Finding the inverse Laplace transform is usually done using a table of Laplace transforms.

First we must write $\bar{y}(s)$ in a form that allows us to use the table. Here we use *partial fractions* to write

$$\bar{y}(s) = \frac{s-1}{(s+1)(s-2)} = \frac{A}{s+1} + \frac{B}{s-2} = \frac{2}{3(s+1)} + \frac{1}{3(s-2)}.$$

Now we can invert $\bar{y}$ using the table, since

$$\mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} = e^{-t}, \quad \mathcal{L}^{-1} \left\{ \frac{1}{s-2} \right\} = e^{2t}.$$

Hence,

$$y(t) = \frac{2}{3}e^{-t} + \frac{1}{3}e^{2t}.$$

Example 2.4.2

A second order ODE is given by

$$\frac{d^2 y}{dt^2} + y = t \quad \text{with} \quad y(0) = 1, \quad y'(0) = 0. \quad (4)$$

- 1 Taking the Laplace transform of both sides of (4),

$$s^2 \bar{y}(s) - y'(0) - sy(0) + \bar{y}(s) = \frac{1}{s^2}.$$

Applying the initial conditions and rearranging leads to

$$(s^2 + 1)\bar{y}(s) = \frac{1}{s^2} + s.$$

- 2 This can now be solved:

$$\bar{y}(s) = \frac{1}{s^2(s^2 + 1)} + \frac{s}{s^2 + 1}.$$

- ③ We want to apply the inverse Laplace transform to $\bar{y}(s)$. The second term appears directly in the table of Laplace transforms. The first term can be dealt with using partial fractions:

$$\bar{y}(s) = \frac{1}{s^2} - \frac{1}{s^2 + 1} + \frac{s}{s^2 + 1}.$$

Now find the inverse Laplace transform using the table:

$$y(t) = t - \sin t + \cos t.$$

We can check that $y(t)$ obeys the ODE and the initial conditions:

$$y'' + y = \sin t - \cos t - t - \sin t + \cos t = t,$$

$$y(0) = \cos(0) = 1, \quad y'(0) = 1 - \cos(0) - \sin(0) = 0.$$

Example 2.4.3

Two coupled ODEs for $x(t)$, $y(t)$ are given by

$$\frac{dx}{dt} + x - 3y = 0, \quad (5)$$

$$\frac{dy}{dt} + 3x - y = e^{-t}, \quad (6)$$

with $x(0) = 0$ and $y(0) = 0$.

Taking the Laplace transform of (5) and (6) will give two simultaneous algebraic equations for $\bar{x}(s)$ and $\bar{y}(s)$.

➊ Doing this, we obtain

$$s\bar{x} + \bar{x} - 3\bar{y} = 0, \quad (7)$$

$$s\bar{y} + 3\bar{x} - \bar{y} = \frac{1}{s+1}. \quad (8)$$

2 Use (7) to eliminate $\bar{x}$ in (8),

$$9\bar{y} + (s-1)(s+1)\bar{y} = 1, \\ \Rightarrow \bar{y} = \frac{1}{9 + (s^2 - 1)} = \frac{1}{s^2 + 8}.$$

3 Invert using the table:

$$y(t) = \frac{1}{\sqrt{8}} \sin(\sqrt{8}t).$$

Then $x(t)$ can be found by rearranging (6):

$$x(t) = \frac{1}{3} \left[y - \frac{dy}{dt} + e^{-t} \right], \\ = \frac{1}{3} \left[\frac{1}{\sqrt{8}} \sin(\sqrt{8}t) - \cos(\sqrt{8}t) + e^{-t} \right]$$

2.5 The Heaviside function and the Dirac delta function

There are two functions for modelling (for example) a discontinuous forcing function or a sudden pulse.

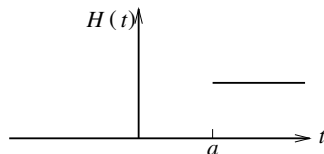
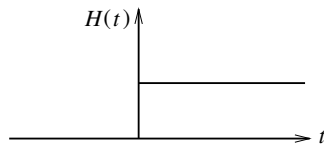
The *Heaviside step function* H is defined by

$$H(t) = \begin{cases} 0 & \text{for } t < 0, \\ 1 & \text{for } t \geq 0. \end{cases}$$

Therefore, $H(t - a)$ is

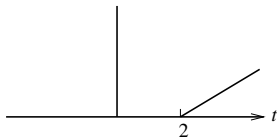
$$H(t - a) = \begin{cases} 0 & \text{for } t < a, \\ 1 & \text{for } t \geq a. \end{cases}$$

So $H(t - a)$ describes, for example, a current which is switched on at $t = a$.



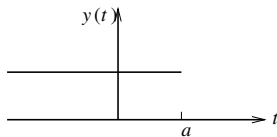
Examples of functions involving H .

$$(i) \quad (t-2)H(t-2) = \begin{cases} 0 & \text{for } t < 2, \\ t-2 & \text{for } t \geq 2. \end{cases}$$



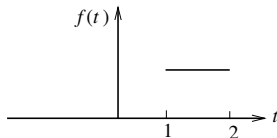
$$(ii) \quad \text{Let } y(t) = 1 - H(t-a). \text{ Then}$$

$$y(t) = \begin{cases} 1 & \text{for } t < a, \\ 0 & \text{for } t \geq a. \end{cases}$$



$$(iii) \quad \text{Let } f(t) = H(t-1) - H(t-2). \text{ Then}$$

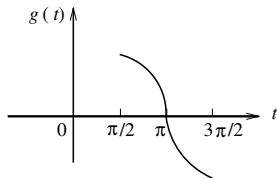
$$f(t) = \begin{cases} 0 & \text{for } t < 1, \\ 1 & \text{for } 1 \leq t < 2, \\ 0 & \text{for } t \geq 2. \end{cases}$$



(iv) Let $g(t) = \sin(t) \left[H\left(t - \frac{\pi}{2}\right) - H\left(t - \frac{3\pi}{2}\right) \right]$.

Then

$$g(t) = \begin{cases} 0 & \text{for } t < \frac{\pi}{2}, \\ \sin t & \text{for } \frac{\pi}{2} \leq t < \frac{3\pi}{2}, \\ 0 & \text{for } t \geq \frac{3\pi}{2}. \end{cases}$$



The Laplace transform of H is VERY USEFUL! It is

$$\begin{aligned}\mathcal{L}\{H(t-a)\} &= \int_0^{\infty} e^{-st} H(t-a) dt, \\ &= \int_a^{\infty} e^{-st} dt, \\ &= \left[-\frac{1}{s} e^{-st} \right]_a^{\infty}, \\ &= \frac{e^{-as}}{s}.\end{aligned}$$

The *Dirac delta function* is defined as

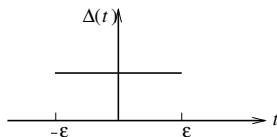
$$\delta(t) = 0 \quad \text{for} \quad t \neq 0.$$

and

$$\int_{-a}^a \delta(t) dt = 1 \quad \text{for any} \quad a > 0.$$

The name “delta function” is slightly misleading. This is not actually a function in the strict mathematical sense. It is something more general, called a *distribution*. As we are more concerned with how to use it, let us just accept the definition above and try to obtain a picture of $\delta(t)$ by considering the function

$$\Delta(t) = \begin{cases} \frac{1}{2\epsilon} & \text{for } -\epsilon < t < \epsilon, \\ 0 & \text{otherwise.} \end{cases}$$



Then

$$\int_{-\infty}^{\infty} \Delta(t) dt = \int_{-\epsilon}^{\epsilon} \frac{1}{2\epsilon} dt = \left[\frac{t}{2\epsilon} \right]_{-\epsilon}^{\epsilon} = 1.$$

So we can formally write

$$\delta(t) = \lim_{\epsilon \rightarrow 0} \Delta(t).$$

Note that this is not a limit in the strict mathematical sense, as the limit does not exist at $t = 0$. But it becomes a correct limit when we integrate it,

$$\int_{-\infty}^{\infty} \delta(t)f(t) dt = \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} \Delta(t)f(t) dt,$$

where f is any continuous function.

We can use the delta function to represent a quantity which occupies a very small region of space, or exists for an instant of time, such as a point force, point charge, or impulse.

An important property of the delta function is that

$$\int_c^d f(t)\delta(t-a) dt = f(a),$$

provided the range of integration includes $t = a$, i.e. $b < a < c$.

The Laplace transform of $\delta(t-a)$ can be found using this result:

$$\mathcal{L}\{\delta(t-a)\} = \int_0^\infty e^{-st}\delta(t-a) dt = e^{-sa},$$

provided $a > 0$.

2.6 Useful Theorems

2.6.1 First Shifting Theorem

The *First Shifting Theorem* is given by

$$\begin{aligned}\mathcal{L}\{e^{at}f(t)\} &= \int_0^{\infty} e^{-st} e^{at} f(t) dt, \\ &= \int_0^{\infty} e^{-(s-a)t} f(t) dt, \\ &= \bar{f}(s-a).\end{aligned}$$

Example 2.6.1

Let $a = 2$, $f(t) = \sin(3t)$ in the First Shifting Theorem.

$$\mathcal{L}\{\sin(3t)\} = \frac{3}{s^2 + 9}, \quad \text{so} \quad \mathcal{L}\{e^{2t} \sin(3t)\} = \frac{3}{(s-2)^2 + 9}.$$

Hence,

$$\mathcal{L}^{-1}\left\{\frac{3}{s^2 - 4s + 13}\right\} = \mathcal{L}^{-1}\left\{\frac{3}{(s-2)^2 + 9}\right\} = e^{2t} \sin(3t).$$

2.6.2 Second Shifting Theorem

We derive the *Second Shifting Theorem* by considering the Laplace transform of $f(t - a)H(t - a)$:

$$\begin{aligned}\mathcal{L}\{f(t - a)H(t - a)\} &= \int_0^{\infty} e^{-st} f(t - a)H(t - a) dt, \\ &= \int_a^{\infty} e^{-st} f(t - a) dt.\end{aligned}$$

Substituting $t = u + a$ gives

$$\begin{aligned}\mathcal{L}\{f(t - a)H(t - a)\} &= \int_0^{\infty} e^{-s(u+a)} f(u) du, \\ &= e^{-as} \int_0^{\infty} e^{-su} f(u) du, \\ &= e^{-as} \bar{f}(s).\end{aligned}$$

Example 2.6.2

Solve the ODE

$$\frac{d^2y}{dt^2} + 9y = 10\delta(t-2),$$

with $y(0) = 0$ and $y'(0) = 1$.

Taking the Laplace transform gives

$$(s^2 + 9)\bar{y} - 1 = 10e^{-2s} \quad \Rightarrow \quad \bar{y} = \frac{1}{s^2 + 9} + \frac{10e^{-2s}}{s^2 + 9}.$$

Now,

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 9} \right\} = \frac{1}{3} \sin(3t),$$

i.e.

$$\mathcal{L}^{-1} \left\{ \frac{10e^{-2s}}{s^2 + 9} \right\} = \frac{10}{3} \sin(3(t-2))H(t-2),$$

where we have used the Second Shifting Theorem. Hence,

$$y(t) = \frac{1}{3} \sin(3t) + \frac{10}{3} \sin(3(t-2))H(t-2).$$

2.6.3 Convolution Theorem

The convolution $f \star g$ of two functions f and g is defined as

$$(f \star g)(t) = \int_0^t f(u)g(t-u) \, du.$$

It can be shown, using double integrals, that the Laplace transform of the convolution is

$$\mathcal{L}\{(f \star g)(t)\} = \bar{f}(s)\bar{g}(s),$$

or alternatively

$$\mathcal{L}^{-1}\{\bar{f}(s)\bar{g}(s)\} = (f \star g)(t) = \int_0^t f(u)g(t-u) \, du.$$

This result is called the *Convolution Theorem*. It is sometimes useful for inverting Laplace transforms where we can identify a product of two functions of s .

Example 2.6.3

Find $\mathcal{L}^{-1} \left\{ \frac{k}{s^2(s^2 + k^2)} \right\}$.

Let $\bar{f}(s) = \frac{1}{s^2}$ and $\bar{g}(s) = \frac{k}{s^2 + k^2}$.

Then $\mathcal{L}^{-1} \left\{ \frac{k}{s^2(s^2 + k^2)} \right\} = \mathcal{L}^{-1} \{ \bar{f}(s)\bar{g}(s) \}$.

From the table of Laplace transforms, we know that $f(t) = t$ and $g(t) = \sin(kt)$, and using the Convolution Theorem

$$\begin{aligned}\mathcal{L}^{-1} \{ \bar{f}(s)\bar{g}(s) \} &= \int_0^t f(u)g(t-u) \, du = \int_0^t u \sin(k(t-u)) \, du, \\ &= \left[-\frac{u \cos(k(t-u))}{-k} \right]_{u=0}^{u=t} - \int_0^t \frac{-\cos(k(t-u))}{-k} \, du, \\ &= \frac{t}{k} - \frac{1}{k} \left[\frac{\sin(k(t-u))}{-k} \right]_{u=0}^{u=t} = \frac{1}{k^2} (kt - \sin(kt)).\end{aligned}$$

2.6.4 Final Value Theorem

The Final Value Theorem states that

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} [s\bar{f}(s)] ,$$

provided $\lim_{t \rightarrow \infty} f(t)$ exists.

This theorem, which is useful in Control Theory, enables the long-time behaviour of a function to be determined from its Laplace transform without the need to find the complete solution.

Further similar results allow for the existence, or non-existence, of the limit of $f(t)$ as $t \rightarrow \infty$ to be determined by inspection of $\bar{f}(s)$.

2.7 Solving ODEs with piecewise elements

ODEs involving the Heaviside function or the delta function are best solved using the Laplace transform.

Example 2.7.1

An oscillator, initially at rest, has constant forcing that is switched off at $t = 2$, i.e.

$$\frac{d^2 y}{dt^2} + y = 1 - H(t - 2),$$

with $y(0) = 0$, $y'(0) = 0$.

Taking the Laplace transform, we obtain

$$s^2 \bar{y} + \bar{y} = \frac{1}{s} - \frac{e^{-2s}}{s} \quad \Rightarrow \quad \bar{y} = \frac{1}{s(s^2 + 1)}(1 - e^{-2s}).$$

Using partial fractions reveals

$$\frac{1}{s(s^2 + 1)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 1} \quad \Rightarrow \quad A = 1, \quad B = -1, \quad C = 0.$$

Hence,

$$\bar{y} = \left(\frac{1}{s} - \frac{s}{s^2 + 1} \right) (1 - e^{-2s}).$$

Now invert $\bar{y}(s)$. The term

$$\frac{1}{s} - \frac{s}{s^2 + 1}$$

can be inverted using the tables, giving $1 - \cos t$. The term

$$\left(\frac{1}{s} - \frac{s}{s^2 + 1} \right) e^{-2s}$$

can be inverted using the Second Shifting Theorem, giving

$$[1 - \cos(t - 2)]H(t - 2).$$

The full solution is

$$y(t) = 1 - \cos t - [1 - \cos(t - 2)]H(t - 2).$$

Since

$$H(t - 2) = \begin{cases} 0 & \text{for } t < 2, \\ 1 & \text{for } t \geq 2, \end{cases}$$

the solution can be written as

$$y(t) = \begin{cases} 1 - \cos t & \text{for } 0 < t < 2, \\ -\cos t + \cos(t - 2) & \text{for } t \geq 2. \end{cases}$$

Again, you can check that y obeys the ODE and the initial conditions (Exercise).